OB+RCT
CATE/Overlap/High dimension/Matching

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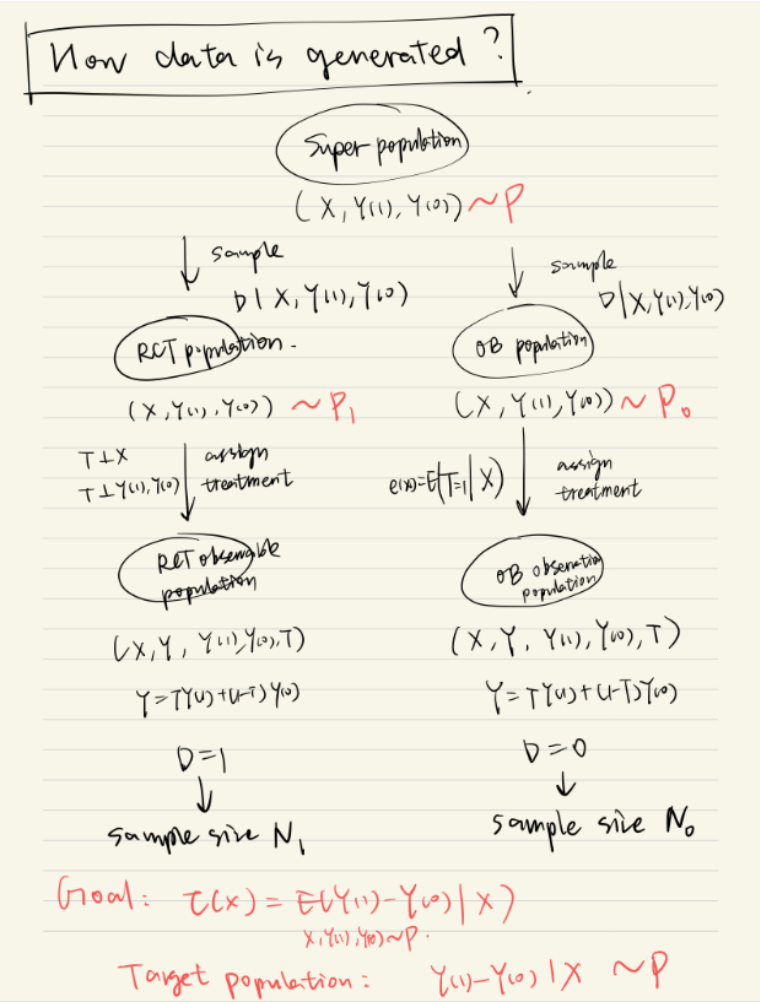
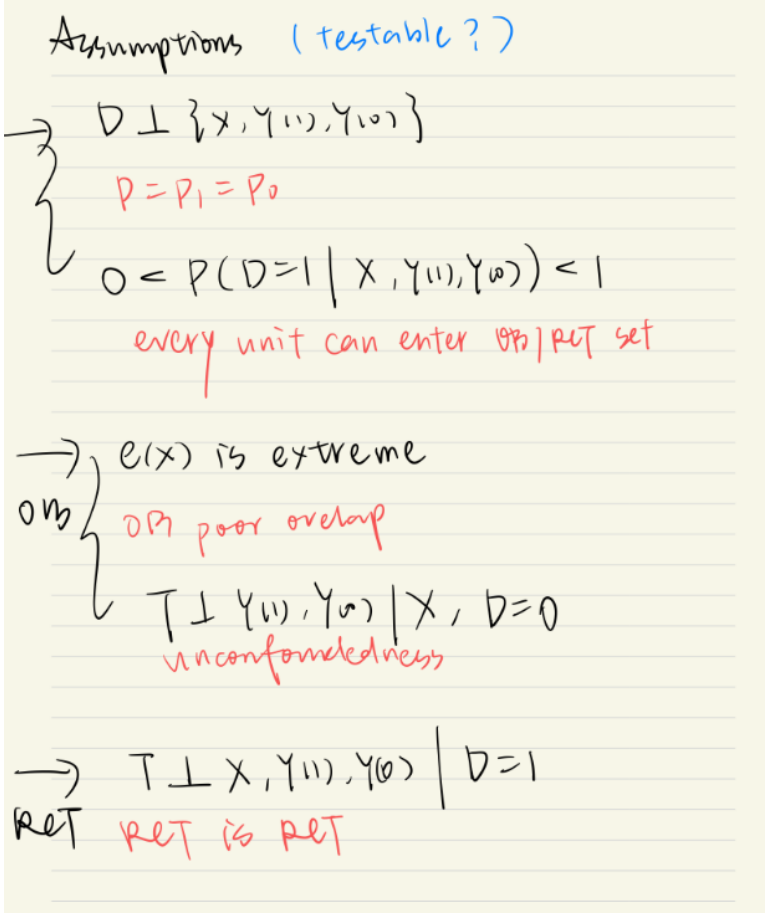
combine rct and ob - why?

- RCT+EC
  - RCT so small
  - RCT may only cover some specific patient(X support region)
- OB+RCT
  - OB with poor overlap(in what sense?) our case
  - OB with unmeasured confounding

combine rct and ob - a data generating view

In super population framework, both RCT and OB are samples from a stochastic distribution. (not finite population and design-based experiment framework) (Liu, L. Thomas, et al. 2025) introduce a double-indexed notation.





OB - overlap assumption and overlap-intuited methods

- Overlap condition influences the performance of estimators (Busso, DiNardo, and McCrary 2014)
- Theoretically, Overlap assumption is a must for the identification of ATE and Strict Overlap is needed for statistical analysis in many papers. (e.g. Causal RF)

? (Crump et al. 2009) [ATE] Characterize(by ps score) optimal subsamples for which the average treatment effect can be estimated most precisely.

we also consider the problem without imposing this restriction. Formally, we choose an estimand $\tau_{S,\mathbb{A}}$ by choosing the set $\mathbb{A} \subset \mathbb{X}$ that minimizes

$$V_{\mathbb{A}}^{\text{eff}} = \frac{1}{\{E(1_{X\in\mathbb{A}})\}^2} E\left[1_{X\in\mathbb{A}}\left\{\frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1-e(X)}\right\}\right].$$

Defining $q(\mathbb{A}) = \text{pr}(X \in \mathbb{A}) = E(1_{X\in\mathbb{A}})$, we can write the objective function as

$$V_{\mathbb{A}}^{\text{eff}} = \frac{1}{q(\mathbb{A})} E\left\{\frac{\sigma_1^2(X)}{e(X)} + \frac{\sigma_0^2(X)}{1-e(X)} \middle| X \in \mathbb{A}\right\}.$$

We seek $\mathbb{A} = \mathbb{A}^*$, which minimizes $V_{\mathbb{A}}^{\text{eff}}$ among all closed subsets $\mathbb{A} \subset \mathbb{X}$.

- (Li, L. E. Thomas, and Li 2019) [ATE] Overlap Weighting

Overlap in IPW

Overlap in PLM

Consider Partial linear model

$$E(Y \mid X, Z) = \beta^T Z + \theta(X)$$

i.e.

$$Y - E(Y|X) = \beta(Z - E(Z|X)) + U, \; E(U|X, Z) = 0$$

In causal settings, given unconfoundedness, $E(Y(1)|X) = \beta + \theta(X)$, $E(Y(0)|X) = \theta(X)$, ATE = β(our estimand).

- (Robinson 1988) (generalized FWL theorem) regression-of-residuals estimator is consistent $\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2 \Phi^{-1})$. if
  - $\Phi = E[(Z - E(Z|X))(Z - E(Z|X))^\top]$ is positive definite.
    - $\Phi^{-1} = \frac{1}{E(e(x)(1-e(x))} = \frac{1}{E(Var(Z|X))}$ (convergence rate & overlap)
  - kernel estimators for first step, other ..

Overlap in R-learner

Considering HTE, now the PLM model becomes:

$$E(Y \mid X, Z) = \beta(X)Z + \theta(X)$$

i.e.

$$Y - m(X) = \beta(X)(Z - e(X)) + U$$

In causal settings, given unconfoundedness, $E(Y(1)|X) = \beta(X) + \theta(X)$, $E(Y(0)|X) = \theta(X)$, CATE $= \beta(X)$ (our estimand).

- (Nie and Wager 2021) equivalently

$$\tau^*(\cdot) = \arg\min_{\tau} \mathbb{E}\left[\left(Y - m^*(X) - (W - e^*(X))\tau(X)\right)^2\right]$$

  therefore estimator(minimize R-loss, cross fitting):

$$\hat{\tau} = \arg\min_{\tau} \left(\hat{L}_n(\tau) + \Lambda_n(\tau)\right)$$

  - penalized kernel R-learner is near oracle(depends on the local robustness property of R-loss) if
    - strict overlap: turning regret bound into MSE bound

$$(1-\eta)^{-2}R(\tau) < E[\{\tau(X_i) - \tau^*(X_i)\}^2] < \eta^{-2}R(\tau)$$

    - RKHS, good estimate of nuisance functions, kernel, ...
  - when $e(X) \approx 0$, R-learner roughly equivalent to X-learner(?)

Overlap in CRF - high-dimensional/heterogeneity

An example of strict overlap in high-dimension case.

- Bias of estimation(Wager and Athey 2018)

$$|\mathbb{E}[\hat{\tau}(x)] - \tau(x)| \lesssim 2Md\left(\frac{\eta s}{2k-1}\right)^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}} = \mathcal{O}\left(s^{-\frac{1}{2}\frac{\log((1-\alpha)^{-1})}{\log(\alpha^{-1})}\frac{\pi}{d}}\right)$$

  - bounded covariates $X_1, \ldots X_s \in [0,1]^d$
  - M-bounded and Lipschitz continuous conditional mean functions
  - strict overlap: for some $\eta > 0$ and all $x \in [0,1]^d$, $\eta < e(x) < 1 - \eta$.
  - A tree predictor grown by recursive partitioning is α-regular
- Tree beats NN under settings of low-dimensional signal structure[1]
- More
  - is strict overlap plausible for tree? how does the balance of data influence the structure(then the performance) of the tree?
  - could match/balance based weighting be used in CRF? (e.g. a balance-aware way of sample weighting/splitting[2])

[1] a data-driven way to determine which nearby observations receive more weight

[2] "An important challenge for future work is to design splitting rules that can automatically choose which characteristic of the training data to split on."

OB - 'overlapness' in population and sample sense

- why poor overlap?
  -
- poor overlap, then?
- when does that matter

OB - overlap & balance

Define P_1 by $\frac{dP_1}{dP} = \frac{e(X)}{E_p(e(X))}$ and P_0 by $\frac{dP_0}{dP} = \frac{1-e(X)}{E_p(1-e(X))}$, which are the distribution of covariates in treatment and control arm. Let $P(Z=1) = \pi$ be a constant $\pi \in (0,1)$.
By Bayes' theorem, we have density ratio(a measure of balance) with propensity ratio(a measure of overlap) as

$$\frac{dP_1}{dP_0} = \frac{P(X|Z=1)}{P(X|Z=0)} = \frac{P(Z=1|X)}{P(Z=0|X)}\frac{P(Z=0)}{P(Z=1)} = \frac{P(Z=1|X)}{P(Z=0|X)}\frac{1-\pi}{\pi}$$

and the mean discrepancy(a measure of balance) is

$$E(X|Z=1) - E(X|Z=0) = E_{P_1}(X) - E_{P_0}(X)$$
$$= E_P(X(\frac{dP_1}{dP} - \frac{dP_0}{dP})) = E_P(X(\frac{e(X)-\pi}{\pi(1-\pi)}))$$

Also can derive other f-divergence form of distance to measure the difference of the two distributions.

$$D_f(Q_1 \| Q_0) := E_{Q_0}[f(\frac{dQ_1}{dQ_0})]$$

OB - strict overlap & balance in high-dimensional setting

high-dimensional: (1) many covariates; (2) complex interactions among covariates in outcome model or propensity model

- Strict overlap with η implies bounded density ratio.

$$b_{min} =: \frac{1-\pi}{\pi}\frac{\eta}{1-\eta} \leq \frac{dP_1}{dP_0} \leq \frac{1-\pi}{\pi}\frac{1-\eta}{\eta} := b_{max}$$

- Strict overlap restricts global discrepancies between P_0 and P_1([support same?]), and these restrictions become more restrictive as the dimension of covariates grows large(D'Amour et al. 2021)[3]

$$D_f(P_1 \| P_0) \leq B_f(b_{min}, b_{max})$$

$$D_f(P_0 \| P_1) \leq B_f(b_{max}^{-1}, b_{min}^{-1})$$

- From data-generating view, if we have high-interaction confounding, then poor high-dimension overlap [?]

[3] where the upper bound B_f is actually free of p, while the left side may not. So strict overlap is very 'strict' in high-dimensional setting.

OB - overlap & balance

Look into the classical setting:

$$e(X) = exp(W_e),\ X \sim N_p(\mu, \Sigma),\ W_e = \beta_0 + \sum_{j=1}^{p} \beta_j X_j \sim N(\beta_0 + \beta^T \mu, \beta^T \Sigma \beta)$$

Does strict overlap hold? for a constant $\eta \in (0, 1)$.
What would influence the divergence $D_f(P_1 || P_0)$?

OB - measure of population overlapness

Overlapness can be a overall concept, or a local concept.

- $min(\eta)$ of strict overlap assumption $\eta < e(X) < 1 - \eta$
- $E(e(x)(1 - e(x))$?
- (Liu and Li 2025) Let $f_z(u) = Pr(e(X) = u \mid Z = z)$, the Bhattacharyya coefficient

$$\phi \equiv \int_0^1 \sqrt{f_1(u) f_0(u)}\, \mathrm{d}u.$$

  - assume propensity function is canonical a logit-normal distribution(by Lyapunov CLT), which can be approximated by Beta distribution
  - (r, ϕ) is sufficient to define this beta dist
  - given r, ϕ is monotonically increasing in both a and b.
- (Kim et al. 2025) Let $\Sigma_z = \mathbb{E}[\phi(x) \otimes \phi(x)\mathbf{1}(Z = z)]$, define weak overlap[?]

$$\Sigma_0 \preceq R\left(\Sigma_1 + \frac{\xi}{n}\mathbf{I}\right) \quad \text{and} \quad \Sigma_1 \preceq R\left(\Sigma_0 + \frac{\xi}{n}\mathbf{I}\right)$$

OB - measure of sample balance

Balance is an overall concept(no local version), and we need a metric to describe the distance of two distributions.

- one-dimensional norm(e,g. overlap region of propensity score[4]); mean-matching/pair-matching (Rubin 1973a)
- Mahalanobis distance[5] (which also contain terms that are functions of the covariates like power/interactions)
- p-values form KS test and paired t-tests (general because the same scale) (Diamond and Sekhon 2013)
- combine ps-score and original X and ..(better firstly orthogonalize by regressing)

[4] for ps score, better to use the linear predictor because it is note close to 0 or 1 (Diamond and Sekhon 2013)

[5] MD does not perform well when covariates have nonellipsoidal distributions?

OB - matching

- (Rubin 1973b) [OB, ATT, one-dimension X, homogeneous/parallel effect]. Use percent reduction in bias as a metric. Matching and regression can reduce bias in observational studies.
- (Diamond and Sekhon 2013) The GenMatch algorithm searches a range of distance metrics to find the particular measure that optimizes post-matching covariate balance. iteratively checking the specification of the propensity score model.

OB - matching and weighting

https://www2.stat.duke.edu/~fl35/teaching/640/Chap3.4_observational_weighting.pdf

Connection to Matching and Regression

- Matching: link "similar" cases in two samples, discard unmatched cases (bottom-up approach)
- Weighting: apply weights to entire samples, designed to create global balance (top-down approach)
- Intrinsic connection: Overlap weighting approaches many-to-many matching as the propensity score model becomes increasingly complex
- Why? The limit of a PS model is a saturated model with a fixed effect (dummy variable) for each design point (or small neighborhood, for continuous variables)

Connection to Matching and Regression

- Let the sample count for x_i in group $z = 0, 1$ be N_{zi}, the estimated PS from the saturated model is $\hat{e}(x_i) = N_{1i}/(N_{0i} + N_{1i})$
- The overlap weight for trt and con units is $N_{0i}/(N_{0i} + N_{1i})$ and $N_{1i}/(N_{0i} + N_{1i})$, respectively
- In many-to-many matching: at each design point x, each trt unit is matched to the N_{0i} controls at the same x, each control to N_{1i} trt units – exactly the same as the overlap weighting
- The total overlap weight for each design point x and hence of $\hat{\tau}(x_i)$ is $N_{0i}N_{1i}/(N_{0i} + N_{1i})$
- This is exactly the precision weight attached to $\bar{y}_{1i} - \bar{y}_{0i}$ in the fixed-effects for each design point OLS outcome model $Y_{zi} = \alpha_i + z\tau + \epsilon_{zi}$

combine rct and ob

- naive-but-possibly-wrong method: just combine the two source, throw it into model
- naive method: separately estimate and average the results.
  - just a linear combination of two estimators, easy to analyze its properties
  - involve tuning(more smart way of tuning?), not end-to-end
- (Kim et al. 2025) Define weak overlap; covariate shift between sources.

Simulations settings

Under what kind of settings would our methods be better than other? N obs data, M rct data; d covariates, within which c are confounders and $d - c$ are prognsotic covariates(assume no instrumental variables)

- $e(x)$, overlap, well-specified or not
- M/N, relative information size of rct sample
- $\mu_0(x), \mu_1(x)$, the response surfaces and CATE $\tau(x)$
- c/d, how many irrelevant covariates
- high-dimensional or not (d/N or c/N, complexity of models)

Simulations settings - how to simulate X

Types of X: xx
(D'Amour et al. 2021) Covariate sequence X is a stochastic process $(X_{(k)})_{(k>0)}$, for any single problem, the investigator selects a finite set of covariates $X_{1:p}$ from the finite pool of covariates $(X_{(k)})_{(k>0)}$.

Simulations settings - linearity of outcome model

What comes along with linear model assumption? What if this is not true?

If the true outcome model is linear, $\mu_{(w)}(x) = x\beta(w)$, then

- the difference of them, i.e. CATE, is linear in X
- fitting one model for each z is equivalent to adding an interaction term Xz
- achieving sample balance[6] just involves balancing the X_i
- If the response surface is nonlinear, all estimates are general biased even if $\bar{x}_1 = \bar{x}_2$. (Rubin 1973b) [ATT, one-dimension]

[6] $\frac{1}{n}\sum_{i=1}^{n}\frac{Z_i\phi_j(X_i)}{e(X_i)} = \frac{1}{n}\sum_{i=1}^{n}\frac{(1-Z_i)\phi_j(X_i)}{1-e(X_i)} = \frac{1}{n}\sum_{i=1}^{n}\phi_j(X_i)$

Some thoughts and questions

- why does overlap influence the performance of estimators in math[7]
- overlap/balance aware adaptive algorithm?
- What is target population? After matching/weighting, would that actually change? (can visualize X)
- After matching/weighting, is the balance better?
- metrics of CATE estimation for simulation? [8]
- poor overlap in high-order-interaction-terms sense? hard to grow a tree? [how much overlap, i.e. how strong tails of $e(x)$]
- (matching-based) data-augmenting then modeling v.s. sample-weighting in DL/Tree models. Would it be different if we double one sampleor weight it by two in model?

[7] 'If the imbalance of the covariates between the two groups is large, the model-based results heavily relies on extrapolation in the region with little overlap, which is sensitive to the model specification assumption.'

[8] (1) $\frac{1}{n}\sum_{i=1}^{n}(\hat{\tau}_i - \tau_i)^2$; (2) $\frac{1}{n}\sum_{i=1}^{n}\hat{\tau}(x_i) - \frac{1}{n}\sum_{i=1}^{n}\tau(x_i)$; (3) $E(\hat{\beta} - \beta)^2$

Summary(before)

- Goal: better estimate CATE
  - data: large-scale Post-OB data and small-scale RCT data
  - problem: poor overlap and severe unbalance
- Method: matching-based
  - data augmentation
  - ITE-imputing loss
- Data source: simulation and real-world data
- Further
  - when would the methods work better(the choice of caliper for matching, the size of ob&rct, the degree of non-overlap, the use of previous algorithm or not, etc)
  - an end-to-end rather than step-wise algorithm
  - multi-treatment

references

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